

Pointed unitary transport between trace-cyclic target representations

MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic

theorem

Source cyclicity and strong shell sums extend the canonical CAR orbit isometry to the whole target.

Statement

Let A be the completed CAR algebra, the completion of the matrix system $M_{2^n}(\mathbb{C})$, and let τ be its normalized trace. Fix a representative shell family: for each chosen pure-GNS class j it specifies an automorphism α_j and partial isometries $w_{j,n}$ between the transported and root shells. Fix the links L_j chosen by this family in the selected atomic representation π . Throughout this Card,

$$T = C^*(\pi(A), \{L_j\}), \quad \iota : A \longrightarrow T, \quad \iota(a) = \pi(a).$$

Thus T is the already constructed concrete unital C^* -algebra, and ι is its injective source map. The links and target are kept fixed. A state means a positive continuous complex linear functional taking 1 to 1. Let $\rho : T \rightarrow B(H)$ and $\sigma : T \rightarrow B(K)$ be unital star representations on complex Hilbert spaces. Suppose $\xi \in H$ and $\eta \in K$ have dense target orbits and realize the same source trace:

$$\langle \xi, \rho(\iota(a))\xi \rangle = \tau(a) = \langle \eta, \sigma(\iota(a))\eta \rangle \quad (a \in A).$$

There exists a unitary $E : H \rightarrow K$ with $E\xi = \eta$ and $E\rho(t) = \sigma(t)E$ for every $t \in T$.

Assumptions

Both Hilbert spaces are complete. The source trace identities and density of the two target orbits are the hypotheses. The vectors automatically have norm 1 by taking $a = 1$. No irreducibility of either representation and no norm separability of T is required.

Conclusion

The resulting unitary is pointed and intertwines the entire fixed target. Source intertwining is obtained first; intertwining the added generators is a separate strong-limit step.

All shell convergence used below is proved inside each actual representation. The argument does not pass strong convergence through an arbitrary star homomorphism. A star on a bounded operator denotes its Hilbert adjoint, corresponding to Lean's postfix dagger on continuous linear operators.

Proof route

Trace estimates kill the residual flag projections on the source-cyclic subspace. Reduction and target cyclicity make this subspace the entire Hilbert space. Pointed CAR transport can then be passed through the shell limits and extended by norm generation.

▼ Proof steps

1. Write f_n for the decreasing root projections, $p_n = f_n - f_{n+1}$, and α_j for the automorphism selected for class j . The shell operators satisfy $w_{j,n}^* w_{j,n} = \alpha_j(p_n)$ and $w_{j,n} w_{j,n}^* = p_n$. The normalized source trace has $\tau(f_n) = 2^{-n}$. Traciality on these supports gives

$$\tau(\alpha_j(f_n)) - \tau(\alpha_j(f_{n+1})) = \tau(p_n).$$

Together with $f_0 = \alpha_j(f_0) = 1$, this yields $\tau(\alpha_j(f_n)) = 2^{-n}$, as recorded by the transported-flag trace theorem.

2. Put $M = \overline{\rho(\iota(A))\xi}$. For every fixed $b \in A$, apply the projection-orbit estimate separately to the root projection f_n and the transported projection $\alpha_j(f_n)$:

$$\begin{aligned} \|\rho(\iota(f_n))\rho(\iota(b))\xi\|^2 &\leq \|b\|^2 \tau(f_n) = \|b\|^2 2^{-n}, \\ \|\rho(\iota(\alpha_j(f_n)))\rho(\iota(b))\xi\|^2 &\leq \|b\|^2 \tau(\alpha_j(f_n)) = \|b\|^2 2^{-n}. \end{aligned}$$

Both right-hand sides tend to zero as $n \rightarrow \infty$. These estimates use the tracial source vector state τ ; no traciality of a state on T is used.

3. Actual-target shell reconstruction provides strong limits S_j, S_j^* of the represented shell and adjoint-shell sums, intersection projections P_j, Q_j for the transported and root flags, and a residual operator R_j , with

$$\rho(g_j) = S_j + R_j, \quad R_j = \rho(g_j)P_j = Q_j R_j P_j,$$

where $g_j = L_j \in T$. For every n , the intersection projection satisfies $P_j \rho(\iota(\alpha_j(f_n))) = P_j$. Hence, for every fixed $b \in A$,

$$\begin{aligned} \|P_j \rho(\iota(b))\xi\| &= \|P_j \rho(\iota(\alpha_j(f_n)))\rho(\iota(b))\xi\| \\ &\leq \|\rho(\iota(\alpha_j(f_n)))\rho(\iota(b))\xi\| \rightarrow 0. \end{aligned}$$

The left-hand side is independent of n , so $P_j \rho(\iota(b))\xi = 0$. Using $Q_j \rho(\iota(f_n)) = Q_j$ and the root-projection estimate gives $Q_j \rho(\iota(b))\xi = 0$ as well. Continuity extends both equalities to M , so $P_j|_M = Q_j|_M = 0$. Therefore $R_j|_M = 0$; taking adjoints of the support relation gives $R_j^* = P_j R_j^* Q_j$ and $R_j^*|_M = 0$.

4. The closed subspace M reduces the source action. Both the shell sums and their adjoints preserve it, so their strong limits preserve it as well. The preceding residual equalities show $\rho(g_j)|_M = S_j|_M$ and $\rho(g_j)^*|_M = S_j^*|_M$. Hence M reduces all generators, and therefore the norm-closed algebra they generate. It contains ξ and thus the entire target orbit. Target cyclicity forces $M = H$. The same argument gives $\overline{\sigma(\iota(A))\eta} = K$. These are the conclusions of `reduces_cyclicSubspace_of_trace` and `denseRange_source_orbit_of_trace_of_cyclic`.

5. For $a, b \in A$, equality of source vector states gives the Gram identity

$$\langle \rho(\iota(a))\xi, \rho(\iota(b))\xi \rangle = \tau(a^*b) = \langle \sigma(\iota(a))\eta, \sigma(\iota(b))\eta \rangle.$$

The pointed cyclic transport theorem applies to the two now-dense source orbits. The orbit isometry extends onto K and yields a unitary $E : H \rightarrow K$ such that $E\xi = \eta$ and $E\rho(\iota(a)) = \sigma(\iota(a))E$ for every $a \in A$.

6. Set $s_{j,N} = \sum_{n < N} w_{j,n}$. Since the source-cyclic spaces are the whole spaces, reconstruction gives, in each representation,

$$\rho(\iota(s_{j,N})) \longrightarrow \rho(g_j), \quad \sigma(\iota(s_{j,N})) \longrightarrow \sigma(g_j)$$

strongly, as well as the corresponding adjoint limits. For $x \in H$, source intertwining at every finite sum reads

$$E\rho(\iota(s_{j,N}))x = \sigma(\iota(s_{j,N}))Ex.$$

Continuity of E and the two displayed limits yield $E\rho(g_j)x = \sigma(g_j)Ex$. This is the limit passage in the linked proof of the stated theorem.

7. Conjugation gives two continuous star homomorphisms $E\rho(\cdot)E^{-1}$ and σ on the same concrete T . They agree on $\iota(A)$ and every g_j , hence on the algebraic star algebra generated by them and then its norm closure. The exact supplier `intertwines_of_source_of_generators` yields $E\rho(t) = \sigma(t)E$ for all $t \in T$.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic`
- The fixed concrete target and source map —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- Trace values on root and transported flags —
`MathlibAnnex.CStarAlgebra.CAR.trace_transportedFlag`
- Projection-orbit estimate for a tracial state —
`MathlibAnnex.Analysis.CStarAlgebra.norm_sq_projection_orbit_le`
- Intersection projections vanish on the source-cyclic space —
`MathlibAnnex.Analysis.CStarAlgebra.projection_eq_zero_on_cyclicSubspace`
- Actual-representation shell reconstruction —
`MathlibAnnex.CStarAlgebra.CAR.exists_targetShellReconstruction`
- Shell and adjoint sums on the source-cyclic space —
`MathlibAnnex.CStarAlgebra.CAR.exists_shell_sums_eq_on_cyclicSubspace`
- Reduction by the source-cyclic space —
`MathlibAnnex.CStarAlgebra.CAR.reduces_cyclicSubspace_of_trace`
- Target cyclicity makes the source-cyclic space full —
`MathlibAnnex.CStarAlgebra.CAR.cyclicSubspace_eq_top_of_trace_of_cyclic`
- Dense source orbit —
`MathlibAnnex.CStarAlgebra.CAR.denseRange_source_orbit_of_trace_of_cyclic`
- Strong sums in each trace-cyclic representation —
`MathlibAnnex.CStarAlgebra.CAR.stronglyConverges_shell_sums_of_trace_of_cyclic`
- Pointed cyclic transport for the common source state —
`StarAlgHom.existsUnique_pointedCyclicTransport`
- Intertwining extends from concrete generators —
`MathlibAnnex.CStarAlgebra.AtomicConstruction.intertwines_of_source_of_generators`

Lean source signature (exact)

```
theorem exists_pointed_unitary_of_trace_of_cyclic
  (family : RepresentativeShellFamily)
  (ρ : Representation (ShellFamilyTarget family) H)
  (σ : Representation (ShellFamilyTarget family) K) (ξ : H) (η : K)
  (hξ : ∀ a, Representation.vectorFunctional
    (ρ.comp (shellFamilySourceHom family)) ξ a = trace a)
  (hη : ∀ a, Representation.vectorFunctional
    (σ.comp (shellFamilySourceHom family)) η a = trace a)
  (hp : DenseRange (fun a ↦ ρ a ξ)) (hσ : DenseRange (fun a ↦ σ a η)) :
  ∃ e : H ≃i[ $\mathbb{C}$ ] K, e ξ = η ∧
  ∀ a, (e : H → L[ $\mathbb{C}$ ] K).comp (ρ a) = (σ a).comp (e : H → L[ $\mathbb{C}$ ] K)
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily</code>	The fixed CAR shell family, once-chosen links, target T and injective source map $\iota : A \rightarrow T$. The normalized source trace is τ .
<code>ρ : Representation (ShellFamilyTarget family) H</code> <code>σ : Representation (...) K</code>	The given unital star representations $\rho : T \rightarrow B(H)$ and $\sigma : T \rightarrow B(K)$, with H, K complete complex Hilbert spaces as in the surrounding source context.
<code>ξ : H</code> ; <code>η : K</code>	The given vectors $\xi \in H$ and $\eta \in K$. Their norms equal one as a consequence of the trace conditions at 1.
<code>hξ : ∀ a, Representation.vectorFunctional</code> <code>(ρ.comp (shellFamilySourceHom family)) ξ a =</code> <code>trace a</code>	For every $a \in A$, $\langle \xi, \rho(\iota(a))\xi \rangle = \tau(a)$; $\cdot$.comp restricts ρ along ι . Inner products are linear in their second argument.
<code>hη : ∀ a, Representation.vectorFunctional</code> <code>(σ.comp (shellFamilySourceHom family)) η a =</code> <code>trace a</code>	For every same source element $a \in A$, $\langle \eta, \sigma(\iota(a))\eta \rangle = \tau(a)$.
<code>hp : DenseRange (fun a ↦ ρ a ξ)</code>	The target orbit $\{\rho(t)\xi : t \in T\}$ is dense in H . Source-orbit density is derived, not this hypothesis.
<code>hσ : DenseRange (fun a ↦ σ a η)</code>	The target orbit $\{\sigma(t)\eta : t \in T\}$ is dense in K . Neither representation is assumed irreducible.
<code>∃ e : H ≃_i[$\mathbb{C}$] K, e ξ = η</code>	One surjective complex-linear isometry $E : H \rightarrow K$ is produced in this direction and sends the specified ξ to the specified η .

In the source	Mathematical meaning
$\forall a, (e : H \rightarrow L[\mathbb{C}] K). \text{comp } (p \ a) = (\sigma \ a). \text{comp } (e : H \rightarrow L[\mathbb{C}] K)$	For every target element $t \in T$ (source a here), that same E satisfies $E\rho(t) = \sigma(t)E$. The casts read the same unitary as a bounded operator. The output intertwines all of T , not only its CAR source.

Further source notes: The linked proof's source restrictions, source-density facts and shell-sum limits are proof-local names; the displayed signature retains the full quantified conclusion.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.exists_pointed_unitary_of_trace_of_cyclic

Accepted content SHA-256: 3b4e2554284d889d714ea72fdcfd034197966df38ab4adda67d2f01fe2eb43a1

Accepted source guide SHA-256: 810c986e8b8b361a5dddc8f1b9dfcca446b107231cd986ceb9dd2ea25b6903d2

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73